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سيكشن: 3

Information Retrieval & Web Search — Lecture 1: Exercise Solution

Given Document Collection:

d1: He likes to wink, he likes to drink
d2: He likes to drink, and drink, and drink
d3: The thing he likes to drink is ink
d4: The ink he likes to drink is pink
d5: He likes to wink, and drink pink ink

1. Term-Document Incidence Matrix

Terms	d1	d2	d3	d4	d5
and	0	1	0	0	1
drink	1	1	1	1	1
he	1	1	1	1	1
ink	0	0	1	1	1
is	0	0	1	1	0
likes	1	1	1	1	1
pink	0	0	0	1	1
the	0	0	1	1	0
thing	0	0	1	0	0
to	1	1	1	1	1
wink	1	0	0	0	1

2. Match the Query: “wink AND drink AND NOT ink”

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Incidence vectors for the query terms (d1 to d5):
• vector(wink) = 1 0 0 0 1
• vector(drink) = 1 1 1 1 1
• vector(ink) = 0 0 1 1 1 → NOT ink = 1 1 0 0 0

Bitwise AND evaluation:
  wink AND drink AND NOT ink
= (10001) AND (11111) AND (11000)
= (10001) AND (11000)
= 10000
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Resulting Incidence Vector: 1 0 0 0 0
Matching Document(s): d1 (“He likes to wink, he likes to drink”)